Task 1: Guided Activity, Course registration

```
import Foundation
var subjects = ["Swift", "UIKit", "SwiftUI"]
subjects.append("Networking")
subjects.insert("Testing", at: 1)
subjects.remove(at: subjects.firstIndex(of: "UIKit")!)

print("Sorted Subjects:", subjects.sorted())

var registered: Set<String> = ["Arun", "Priya", "Kumar"]
registered.insert("Priya")
registered.insert("Priya")
print("Registered Count:", registered.count)

var marks: [String: Int] = [
    "Arun": 85,
    "Priya": 92,
    "Kumar": 78
]

marks["Divya"] = 88
marks["Arun"] = 90
print("Unknown Student Mark:", marks["Ravi"] as Any)

print("Student Marks:")
for name in marks.keys.sorted() {
    print("\(name): \(marks[name]!)")
}

if let highest = marks.max(by: { $0.value < $1.value }) {
    print("Highest Mark:", highest.value)
    print("Student Name:", highest.key)
}
```

OUTPUT:

```
Sorted Subjects: ["Networking", "Swift", "SwiftUI", "Testing"]
Registered Count: 3
Unknown Student Mark: nil
Student Marks:
Arun: 90
Divya: 88
Kumar: 78
Priya: 92
Highest Mark: 92
Student Name: Priya
```

Task 2: Practice Task, Word frequency counter

```
import Foundation

let sentence = "swift is fun and swift is fast"
let words = sentence.split(separator: " ").map { String($0) }

print("Number of words:", words.count)

let uniqueWords = Set(words)
print("Unique words:", uniqueWords)

var frequency: [String: Int] = [:]
for word in words {
    frequency[word, default: 0] += 1
}
print("Word Frequencies:")
for word in frequency.keys.sorted() {
    print("\(word): \(frequency[word]!)")
}

let rankedWords = frequency.sorted {
    if $0.value != $1.value {
        return $0.value > $1.value
    }
    return $0.key < $1.key
}
print("Words Ranked by Frequency:")
for (word, count) in rankedWords {
    print("\(word): \(count)")
}
```

OUTPUT:

```
Number of words: 7
Unique words: ["fun", "swift", "is", "and", "fast"]
Word Frequencies:
and: 1
fast: 1
fun: 1
is: 2
swift: 2
Words Ranked by Frequency:
is: 2
swift: 2
and: 1
fast: 1
fun: 1
```

Task 3: Challenge Task, Set algebra and de-duplication

```
import Foundation

let swiftClub: Set<String> = ["Anu", "Ben", "Chitra", "Dev"]
let uiClub: Set<String> = ["Chitra", "Dev", "Esha"]

let bothClubs = swiftClub.intersection(uiClub)
print("Members in both clubs:", bothClubs.sorted())

let onlySwift = swiftClub.subtracting(uiClub)
print("Only in Swift club:", onlySwift.sorted())

let everyone = swiftClub.union(uiClub)
print("Everyone:", everyone.sorted())

let exactlyOne = swiftClub.symmetricDifference(uiClub)
print("Members in exactly one club:", exactlyOne.sorted())

let numbers = [3, 1, 3, 2, 1, 4]
var uniqueNumbers: [Int] = []
for number in numbers {
    if !uniqueNumbers.contains(number) {
        uniqueNumbers.append(number)
    }
}
print("Unique numbers in original order:", uniqueNumbers)
```

OUTPUT:

```
Members in both clubs: ["Chitra", "Dev"]
Only in Swift club: ["Anu", "Ben"]
Everyone: ["Anu", "Ben", "Chitra", "Dev", "Esha"]
Members in exactly one club: ["Anu", "Ben", "Esha"]
Unique numbers in original order: [3, 1, 2, 4]
```